Scalability & Future Plan

Date	29 October 2026	Maximum Marks	1 Mark
Team ID	EV-BI-04	Project Name	EV Market Performance Insights Dashboard

Scalability & Future Plan: This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Static CSV Dataset	Manual file updates required for fresh market data.	High
2	Tableau Public Public Load Limit	Potential query lag under heavy concurrent user requests.	Medium
3	Absence of Predictive Analytics	Limited to historical insights without automated sales forecasting.	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	**User Load**	Single web server (Flask local/free tier).	Deploy Flask on AWS EC2 / Heroku with Auto-Scaling & CDN caching.
2	**Data Storage**	Local CSV File (`ev_market_train.csv`).	Migrate data to PostgreSQL / Snowflake Data Warehouse.
3	**Performance**	Direct client-side embedded rendering.	Implement Redis caching for static dashboard embeds and API calls.
4	**Security**	Open portal without role control.	Add OAuth2 authentication & Row-Level Security (RLS) in Tableau Cloud.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Live API data integration & automated daily ETL refresh pipeline.	Q1 2027	Real-time market insights.
Phase 3	ML-based sales forecasting & battery degradation prediction models.	Q2 2027	Predictive BI capabilities.
Phase 4	Global charging station GIS map integration & mobile app launch.	Q3 2027	Complete EV ecosystem view.